NETWORK COMMAND-LINE TOOLS ASSIGNMENT

IP configuration • Connectivity testing • Route tracing • DNS lookup • Troubleshooting

1. Find the Computer's IPv4 Address and Default Gateway Using ipconfig

Objective: Use the Windows Command Prompt to display the network configuration and identify the computer's IPv4 address and default gateway.

Procedure

1. Open the Start menu, type cmd, and open Command Prompt.
2. Enter ipconfig and press Enter. The command displays basic information for active network adapters.
3. Locate the active Wi-Fi or Ethernet adapter. Read the IPv4 Address and Default Gateway fields.
4. For more detailed information, run ipconfig /all. This also displays the subnet mask, DNS servers, DHCP status and physical address.

Example command output

```
C:\> ipconfig
Wireless LAN adapter Wi-Fi:
   IPv4 Address. . . . . . . . . . . : 192.168.1.105
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
```

Details identified

Parameter	Example value	Meaning
IPv4 address	192.168.1.105	Unique address assigned to this computer on the local network.
Default gateway	192.168.1.1	Router address used to send traffic to other networks, including the Internet.
Subnet mask	255.255.255.0	Identifies the network portion and host portion of the IPv4 address.

Result: The example computer has IPv4 address 192.168.1.105 and uses 192.168.1.1 as its default gateway. Actual values depend on the local router and network. A screenshot should show the Command Prompt output with these fields visible.

Screenshot evidence: Capture the ipconfig output on your own computer. The example values above are illustrative and must not be represented as a live reading from your device.

2. Test Connectivity to www.google.com Using ping

Objective: Use ping to test whether the computer can reach Google's website and calculate the average round-trip response time reported by the command.

Procedure

5. Open Command Prompt.
6. Run ping www.google.com and press Enter.
7. Windows sends four ICMP Echo Requests by default and displays the replies, if received.
8. At the end of the output, read the statistics and note the Average time value. If the command reports packet loss, record that as well.

Example command and illustrative output

```
C:\> ping www.google.com
Pinging www.google.com [142.250.183.100] with 32 bytes of data:
Reply from 142.250.183.100: bytes=32 time=24ms TTL=117
Reply from 142.250.183.100: bytes=32 time=26ms TTL=117
Reply from 142.250.183.100: bytes=32 time=23ms TTL=117
Reply from 142.250.183.100: bytes=32 time=27ms TTL=117
Ping statistics for 142.250.183.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 27ms, Average = 25ms
```

Result and interpretation

Illustrative average response time: 25 milliseconds (ms), with 0% packet loss in this sample. This means the four test packets received replies and the average round-trip time was 25 ms. Your actual average may be different depending on your Internet connection, routing, server location and network congestion.

If the output shows Request timed out, it may indicate packet loss or that ICMP traffic is filtered. A failed ping does not always mean the website is unavailable because some networks block ping requests.

Screenshot evidence: Capture the complete ping command output, including the packet statistics and average response time.

3. Trace the Route to www.flipkart.com Using tracert

Objective: Use tracert to display the network hops between the computer and the destination hostname and determine how many hops are shown before the destination is reached.

Procedure

9. Open Command Prompt.
10. Run tracert www.flipkart.com and press Enter.
11. Wait for the trace to complete. Each numbered row represents a router or network hop along the route.
12. Count the numbered hop rows. If the destination is reached, the final responding hop identifies the destination path. Asterisks indicate that a probe did not receive a response within the timeout; they do not necessarily mean the route is broken.

Example command output (illustrative)

```
C:\> tracert www.flipkart.com
Tracing route to www.flipkart.com [203.0.113.25]
over a maximum of 30 hops:
  1     2 ms     1 ms     2 ms  192.168.1.1
  2     8 ms     7 ms     8 ms  10.20.0.1
  3    14 ms    13 ms    15 ms  198.51.100.1
  4    22 ms    21 ms    23 ms  203.0.113.25
Trace complete.
```

Result and interpretation

Illustrative result: The trace reaches the destination in 4 hops. A hop is one intermediate routing step between the source and destination. The number of hops and router addresses will vary by ISP, location, destination infrastructure and time of testing.

Note: The IP addresses shown in this sample are reserved documentation addresses and are not actual Flipkart server addresses. In a real test, record the hop count from your own tracert output. If the trace contains asterisks, count the numbered rows and report whether the destination was actually reached.

Screenshot evidence: Capture the tracert command and its final Trace complete message, if present.

4. Find the IP Address of www.spotify.com Using nslookup

Objective: Use nslookup to query the Domain Name System (DNS) and display the IP address associated with Spotify's hostname.

Procedure

13. Open Command Prompt.
14. Type nslookup www.spotify.com and press Enter.
15. Read the Server and Address lines to identify the DNS resolver that answered the query.
16. Read the Name and Address or Addresses lines in the answer section to identify the IP address returned for the hostname. DNS may return multiple addresses or different results over time.

Example command output (illustrative)

```
C:\> nslookup www.spotify.com
Server:  router.local
Address:  192.168.1.1

Non-authoritative answer:
Name:    www.spotify.com
Addresses:  104.199.65.124
```

Result and one-line explanation

Illustrative DNS result: www.spotify.com resolves to 104.199.65.124 in this sample. The returned address can differ because DNS responses may vary by resolver, location, load balancing and time.

One-line explanation: nslookup verifies that DNS can translate a hostname into an IP address and shows which DNS server supplied the response.

Screenshot evidence: Capture the nslookup command output showing the queried hostname and returned address.

5. Troubleshoot a Friend's Inability to Access Zomato

Scenario: A friend cannot open Zomato's website on a laptop. The aim is to determine whether the problem is caused by local IP configuration, DNS resolution or the network route.

Three-command troubleshooting sequence

Order and command	What it checks	How to interpret the result
1. ipconfig /all	Checks the laptop's IPv4 address, subnet mask, default gateway and DNS server settings.	An APIPA address (169.254.x.x), missing gateway or incorrect DNS server suggests a local configuration or DHCP problem.
2. ping www.zomato.com	Tests whether the hostname resolves and whether the destination responds to ICMP.	If the name cannot be resolved, investigate DNS. If it resolves but times out, check connectivity or possible ICMP filtering.
3. tracert www.zomato.com	Shows the route taken toward the destination and where responses stop.	A trace that stops near the laptop/router may indicate a local gateway or ISP routing issue. Asterisks can also mean routers do not reply to trace probes.

Explanation of the sequence

First, ipconfig /all checks whether the laptop has valid network settings. Second, ping www.zomato.com tests both hostname resolution and basic reachability. Third, tracert www.zomato.com helps locate the point along the route where communication may be failing.

Possible next actions based on findings

- If the laptop has an invalid address or no default gateway, reconnect to Wi-Fi or Ethernet and renew the DHCP lease using ipconfig /release followed by ipconfig /renew.
- If the DNS server setting appears incorrect or the hostname does not resolve, test with nslookup www.zomato.com and correct the DNS configuration if required.
- If the route fails immediately after the local gateway, restart or troubleshoot the router and check the ISP connection.
- If ping and tracert fail but the site opens in a browser, ICMP may be filtered. Test the browser, another website and the network connection before concluding that Zomato is down.

Conclusion

The three selected commands provide a logical first diagnosis: ipconfig checks local addressing, ping checks name resolution and basic reachability, and tracert examines the path to the destination. Further testing with nslookup can isolate DNS problems when required.

Practical evidence checklist

- Screenshot of ipconfig showing the actual IPv4 address and default gateway.
- Screenshot of ping www.google.com showing replies, packet loss and average response time.
- Screenshot of tracert www.flipkart.com showing the hop rows and whether the trace completed.
- Screenshot of nslookup www.spotify.com showing the DNS response.
- For the Zomato scenario, include the three commands and a brief note describing what each result indicates.

Important note: The sample IP addresses, ping times and hop count in this document are illustrative examples, not live measurements from your computer. DNS answers and routes change. Run the commands on your own device and replace the sample observations with your actual results before submitting practical evidence.